## C3.3 Types of chemical reactions

### Summary

Chemical reactions can be classified according to changes at the atomic and molecular level. Examples of these include reduction, oxidation and neutralisation reactions.

### Underlying knowledge and understanding

Learners should be familiar with combustion, thermal decomposition, oxidation and displacement reactions. They will be familiar with defining acids and alkalis in terms of neutralisation reactions. Learners will have met reactions of acids with alkalis to produce a salt and water and reactions of acids with metals to produce a salt and hydrogen. They should have met the pH scale for measuring acidity and alkalinity, and some indicators.

### Common misconceptions

Learners commonly intuitively adhere to the idea that hydrogen ions in an acid are still part of the molecule, not free in the solution. They tend to have little understanding of pH, for example, they tend to think that alkalis are less corrosiv than acids. Learners also may think that the strength of acids and bases and concentration mean the same thing.

### Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

| Reference | Mathematical learning outcomes | Mathematical skills |
|---|---|---|
| CM3.3i | arithmetic computation, ratio, percentage and multistep calculations permeates quantitative chemistry | M1a, M1c, M1d |

| Topic content | | | Opportunities to cover: | | Practical suggestions |
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| **Learning outcomes** | | **To include** | **Maths** | **Working scientifically** | |
| C3.3a | explain reduction and oxidation in terms of loss or gain of oxygen, identifying which species are oxidised and which are reduced | the concept of oxidising agent and reducing agent | | WS1.4a | |
| **C3.3b** | **explain reduction and oxidation in terms of gain or loss of electrons, identifying which species are oxidised and which are reduced** | | | WS1.4a | |
| C3.3c | recall that acids form hydrogen ions when they dissolve in water and solutions of alkalis contain hydroxide ions | | | WS1.4a | |
| C3.3d | describe neutralisation as acid reacting with alkali or a base to form a salt plus water | | | WS1.4a | Production of pure dry sample of salt. (PAG C7) |
| C3.3e | recognise that aqueous neutralisation reactions can be generalised to hydrogen ions reacting with hydroxide ions to form water | | | WS1.4a | |
| C3.3f | recall that carbonates and some metals react with acids and write balanced equations predicting products from given reactants | | | WS1.4a | |
| **C3.3g** | **use and explain the terms dilute and concentrated (amount of substance) and weak and strong (degree of ionisation) in relation to acids** | **ratio of amount of acid to volume of solution** | M1a, M1c, M1d | WS1.4a | |
| C3.3h | recall that relative acidity and alkalinity are measured by pH | | | WS1.4a | |

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|---|---|---|---|---|---|
| **C3.3i** | **describe neutrality and relative acidity and alkalinity in terms of the effect of the concentration of hydrogen ions on the numerical value of pH (whole numbers only)** | **pH of titration curves** | | WS1.4a | Neutralisation reactions. (PAG C6) |
| **C3.3j** | **use the idea that as hydrogen ion concentration increases by a factor of ten, the pH value of a solution decreases by one** | | M1a, M1c, M1d | WS1.4a | |
| C3.3k | describe techniques and apparatus used to measure pH | the use of universal indicator and pH meters | | | Determining pH of unknown solutions. (PAG C6)<br>Use of pH probes. (PAG C6) |